

$^{33}\text{S}(\alpha, \text{d})$ [1975Na18, 1977Na10](#)

$J^\pi=3/2^+$ for ^{33}S ground state.

[1975Na18, 1977Na10](#): a 40-MeV α beam was produced from the Michigan State University cyclotron. Targets were 76.8%-enriched ^{33}S of $15 \mu\text{g}/\text{cm}^2$ sandwiched between layers of Formvar and carbon foils. Reaction products were detected in the focal plane of an Enge split-pole magnetic spectrograph with a proportional-counter plastic-scintillator combination with FWHM=40-60 keV. Measured $\sigma(E_d, \theta)$. Deduced levels, J, π , and L-transfers.

 ^{35}Cl Levels

<u>E(level)[†]</u>	<u>L[†]</u>	<u>Comments</u>
6200 <i>10</i>	6	
7170 <i>10</i>	6	
7670 <i>10</i>	6	
7750 <i>10</i>	6	
7870 <i>10</i>	4+6	L: 1977Na10 states that this is excited by a mixture of L=4 and 6 transfers. The L=6 admixture can be attributed to a $(f_{7/2})^2_{7,0}$ transfer whereas the L=4 component is associated with a mixture of $(f_{7/2})^2_{5,0}$ and $(f_{7/2}, p_{3/2})_{5,0}$ transfers.
8010 <i>10</i>	6	
8100 <i>10</i>	6	
8700 <i>10</i>	6	
8840 <i>10</i>	6	
9150 <i>10</i>	6	
9450 <i>10</i>	6	

[†] From [1977Na10](#). Experimental angular distributions of L=4, 5, 6 to ^{42}Sc and ^{34}Cl levels with known J^π are compared with the $\sigma(E_d, \theta)$ measured in [1977Na10](#) to assign L-transfers.